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St. George's University
Grenada, West Indies

Cellular Biology of Vesicular Transport
Dr. Mai

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Objectives

SOM.MK.1.BPM1.1.FTM.3.HCB.0687	List the two major types of vesicular transport.
SOM.MK.1.BPM1.1.FTM.3.HCB.0688	Describe the precise targeting of vesicles within the cell.
SOM.MK.1.BPM1.1.FTM.3.HCB.0689	Describe the three different mechanisms of endocytosis.
SOM.MK.1.BPM1.1.FTM.3.HCB.0690	Describe the role of endosomes in vesicular transport.
SOM.MK.1.BPM1.1.FTM.3.HCB.0691	Describe the four pathways for processing internalized ligand-receptor complexes.
SOM.MK.1.BPM1.1.FTM.3.HCB.0692	Describe the two general pathways of exocytosis.

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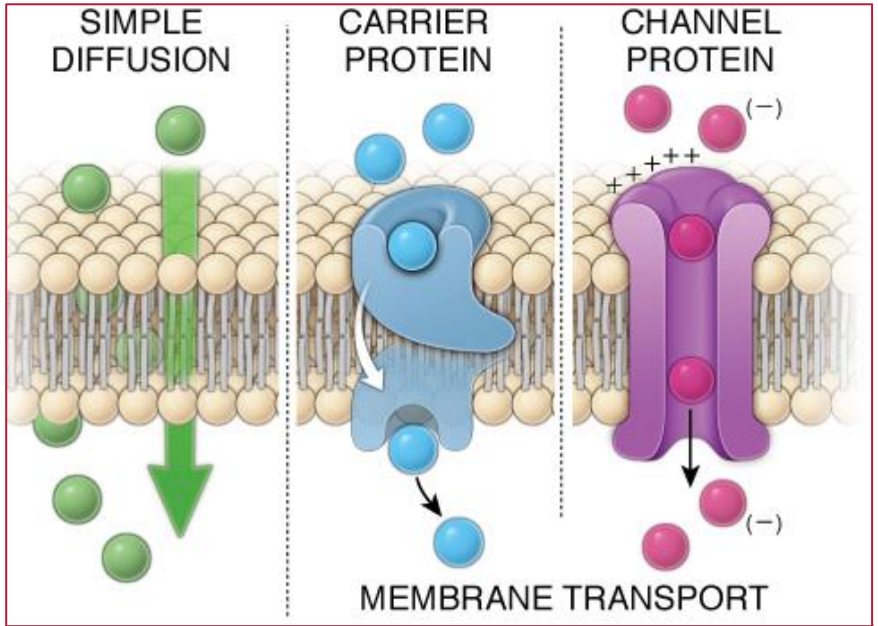
Membrane Transport

- **Some substances cross the membrane by simple diffusion down the concentration gradient**
  - Small, hydrophobic, non-polar, uncharged molecules and gasses, e.g. O_2, CO_2
  - No energy requirements
  - No proteins involved
- **All other molecules require membrane transport proteins**
  - **Carrier proteins**
    - Small water-soluble molecules
    - Protein undergoes conformational changes after binging to the molecule
    - May be active (require energy) e.g. Na^+/K^+
    - Or passive (no energy requirement) e.g. glucose
  - **Channel proteins**
    - Small water-soluble molecules
    - Transmembrane proteins
    - Contains a pore domain that creates ion-selective channels
    - Regulated in different ways
      - Voltage-gated
      - Ligand-gated
      - Mechanical gated



Substances that enter or leave the cell must pass through the plasma membrane

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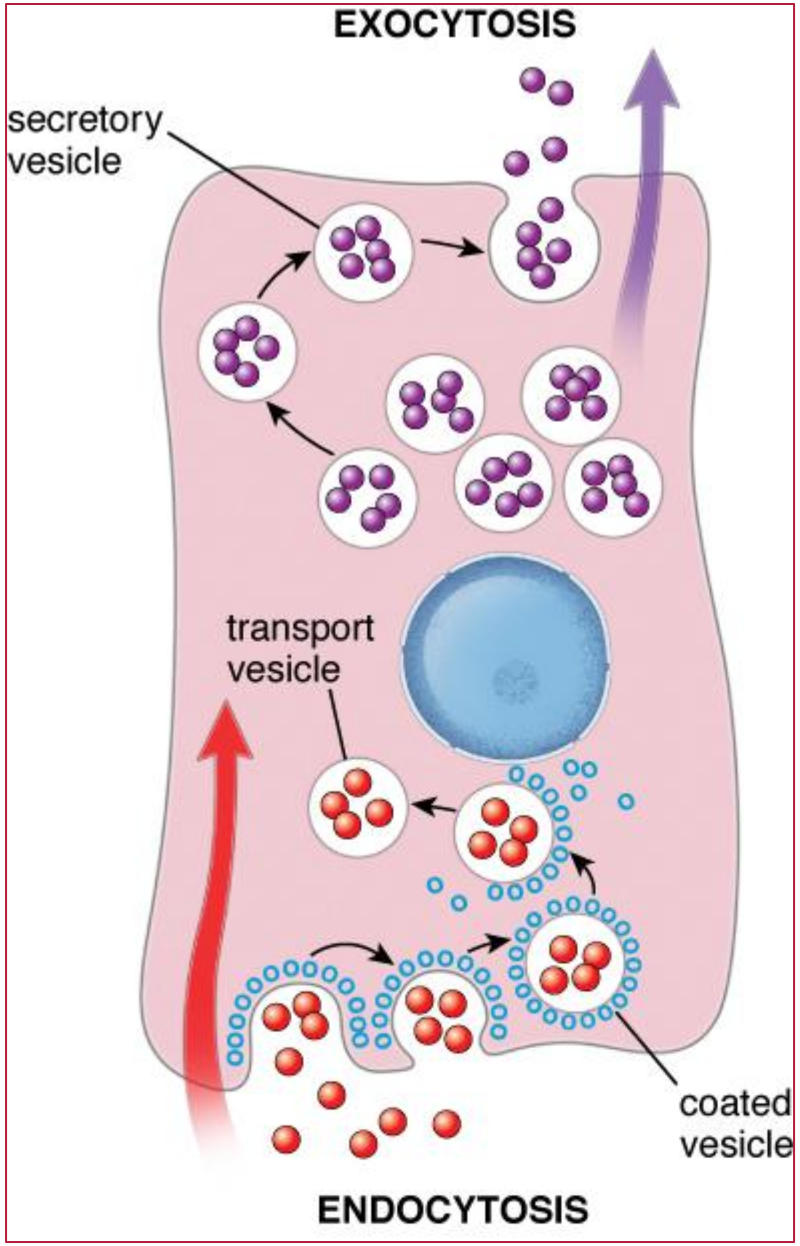
Vesicular transport

- Some substances enter and leave cells by vesicular transport
- Involves configurational changes in the plasma membrane
- Maintains the integrity of the plasma membrane
- Mechanism by which large molecules **enter, leave and move within** the cell **(vesicle budding)**
  - Movement between the ER and Golgi is an example
- Vesicular transport by which substances **eNter** the cell **(ENdocytosis)**
  - Controls the composition of the plasma membrane and cellular response to changes in the external environment
  - Nutrient uptake, cell signaling and shape changes
- Vesicular transport by which substances **eXit** the cell **(EXocytosis)**
- Two pathways
  - Constitutive – continuous
  - Regulated – signal dependent



Exocytosis and endocytosis are coupled: if exocytosis is abolished, no endocytosis takes place.

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Endocytosis

- Uptake of membrane proteins, fluids, nutrients, lipids, and signaling molecules from the extracellular environment into the cell by endocytic vesicles
- Can be classified as clathrin-dependent or clathrin-independent
- Three major mechanisms
    - **Pinocytosis**
        - Non-specific
        - Clathrin-independent
    - **Phagocytosis**
        - Clathrin-independent
        - Actin-dependent
    - **Receptor-mediated**
        - Cargo-specific
        - Clathrin-dependent

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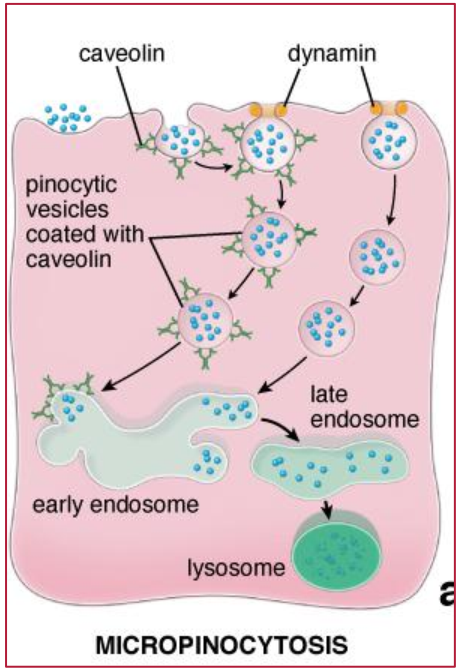
Endocytosis – pinocytosis (*“cell drinking”*)

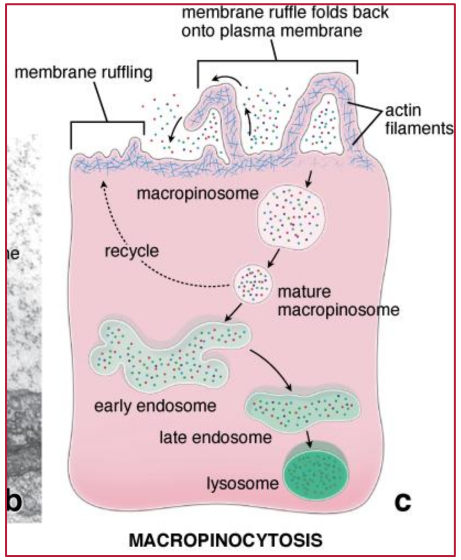
Micropinocytosis

- Non-specific ingestion of fluid and small protein molecules via small vesicles
  - Almost all cells
  - Continuous **(constitutive pathway)**
  - Vesicle formation is associated with **caveolin** and **flotillin** proteins **(lipid raft associated)**
  - **Clathrin-independent**



Macropinocytosis

- Non-specific uptake of extracellular fluids, solutes, nutrients and antigens.
- Regulated
- **Clathrin-independent**
- Requires rearrangement of the actin cytoskeleton **(actin-dependent)** which causes **surface membrane ruffles**
  - become elongated and folds back onto the plasma membrane to trap the fluid
  - Macropinosomes
  - Specifically employed by immune cells to sample their environment



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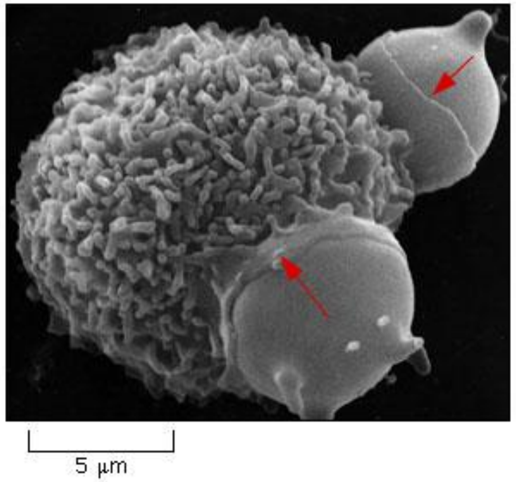
Endocytosis - phagocytosis

- Ingestion of large particles
  - Microorganisms
  - Apoptotic cell debris
  - Non-biological materials
- Generally performed by **specialized phagocytes**
  - Macrophage
  - Neutrophil
- **Receptor mediated**
  - Recognition of pathogen-associated molecular patterns (PAMPs)
  - Particle binds to plasma membrane receptor - Ex. Antibodies
- Extension of pseudopods **(Actin-dependent)**
  - Dependent on Actin microfilament polymerization
- **Clathrin-independent**



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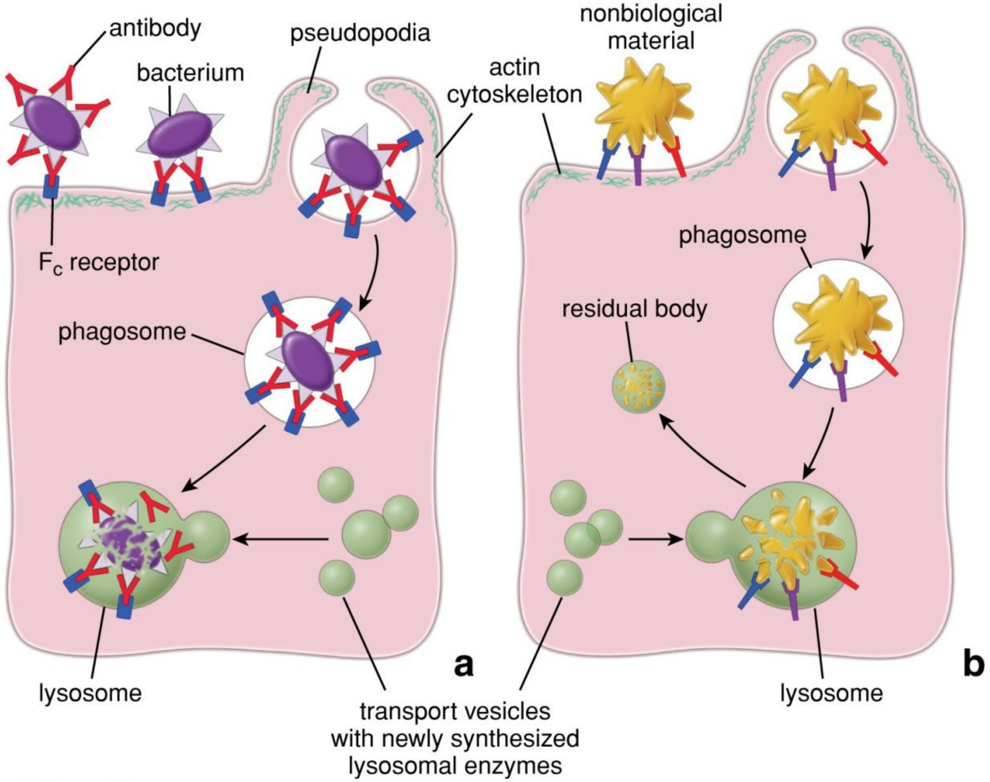

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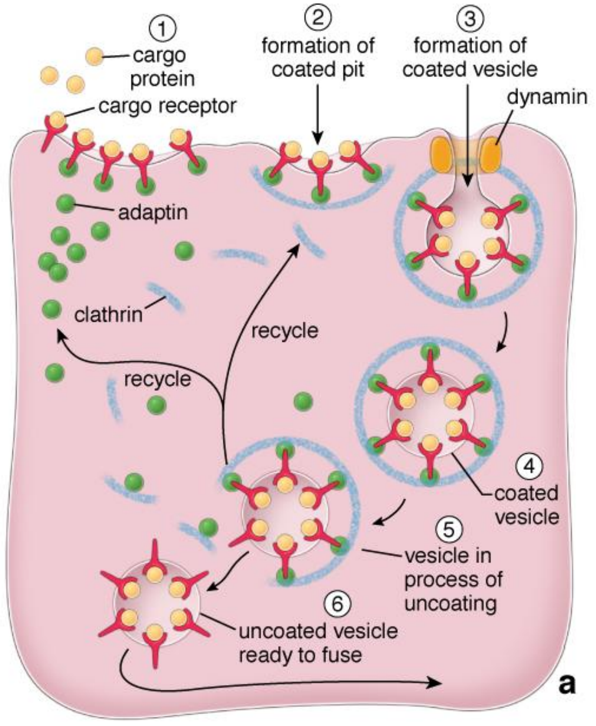
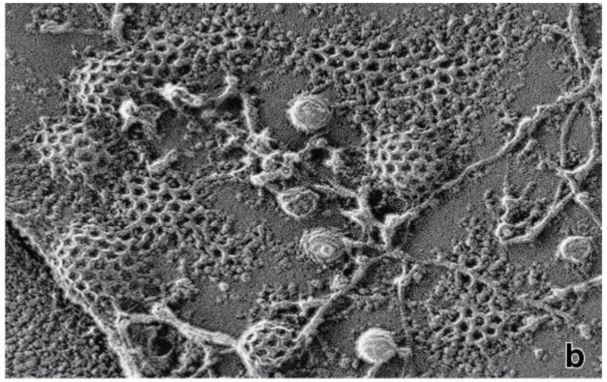
Endocytosis – Receptor mediated

- Entry of specific molecules
- Cargo receptors accumulate in well-defined regions of the cell membrane **(lipid rafts)**
  - Aggregation of **clathrin molecules** on the cytoplasmic surface of membrane **- Coated pits**
  - Recognize and bind to specific molecules
  - Clathrin assemble into a basket-like cage that change the shape of the membrane pulling the **cargo** into the lumen of a forming vesicle
  - **Dynamin** a mechanoenzyme mediates the pinching off from the plasma membrane
- Results in a coated vesicle – **clathrin-dependent endocytosis**





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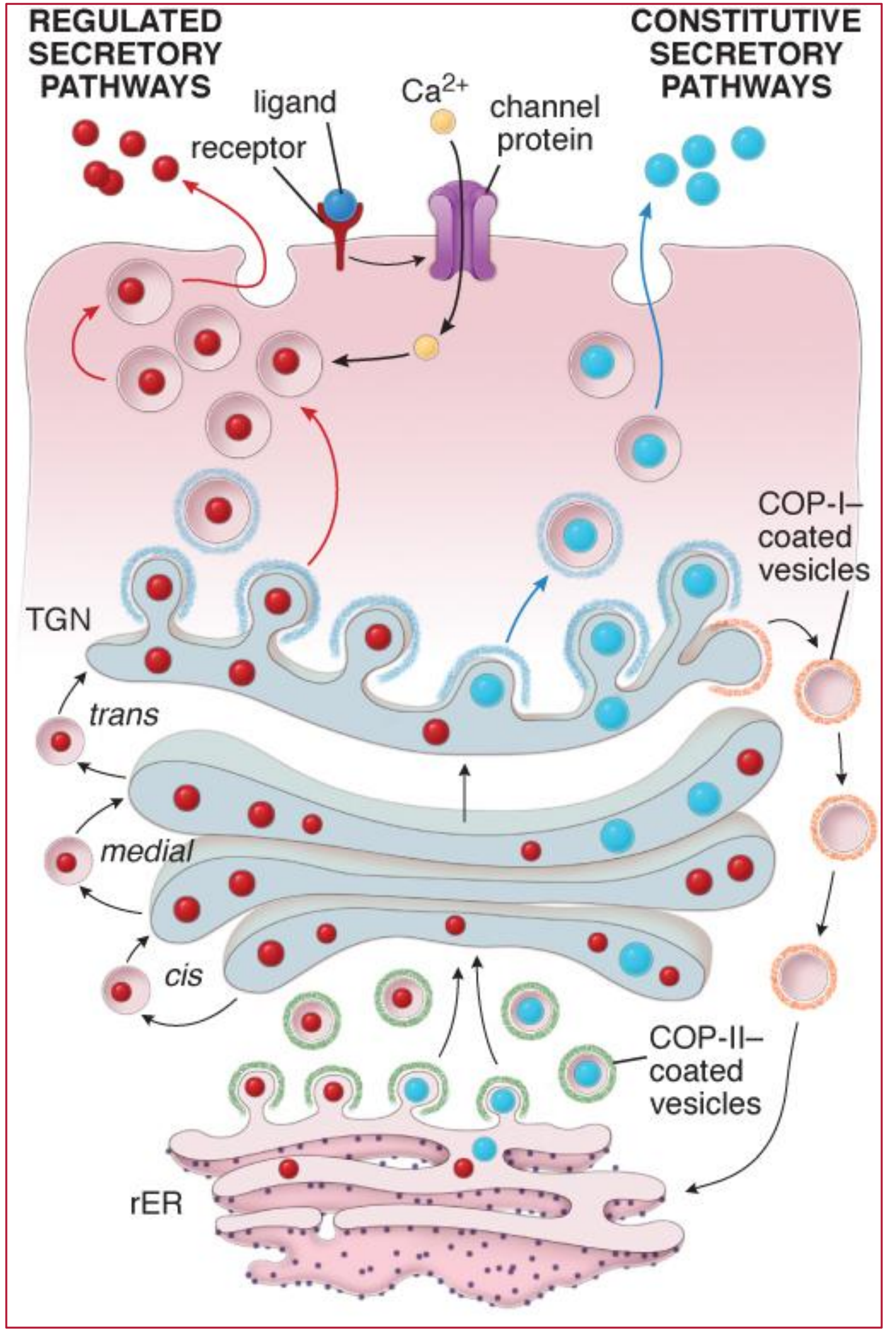
Exocytosis

- The process by which a vesicle, produced by the Golgi apparatus, **moves from the cytoplasm to the plasma membrane**
  - Here it fuses to and discharges its contents into the extracellular space
  - The membrane portion is then recycled during endocytosis
- There are **two pathways**
  - **Constitutive** – a continuous process
    - Present in some degree in all cells
    - Proteins that leave the cell via this pathway are **secreted immediately** after their synthesis
    - e.g. antibodies by plasma cells and procollagen by fibroblasts
  - **Regulated secretory pathway**
    - Specialized cells concentrate secretory proteins and **temporarily store** them in secretory vesicles in the cytoplasm
    - Endocrine, exocrine cells and neurons
    - A regulatory event is required for secretion to occur
    - This stimulates the vesicles to move to and fuse with the plasma membrane



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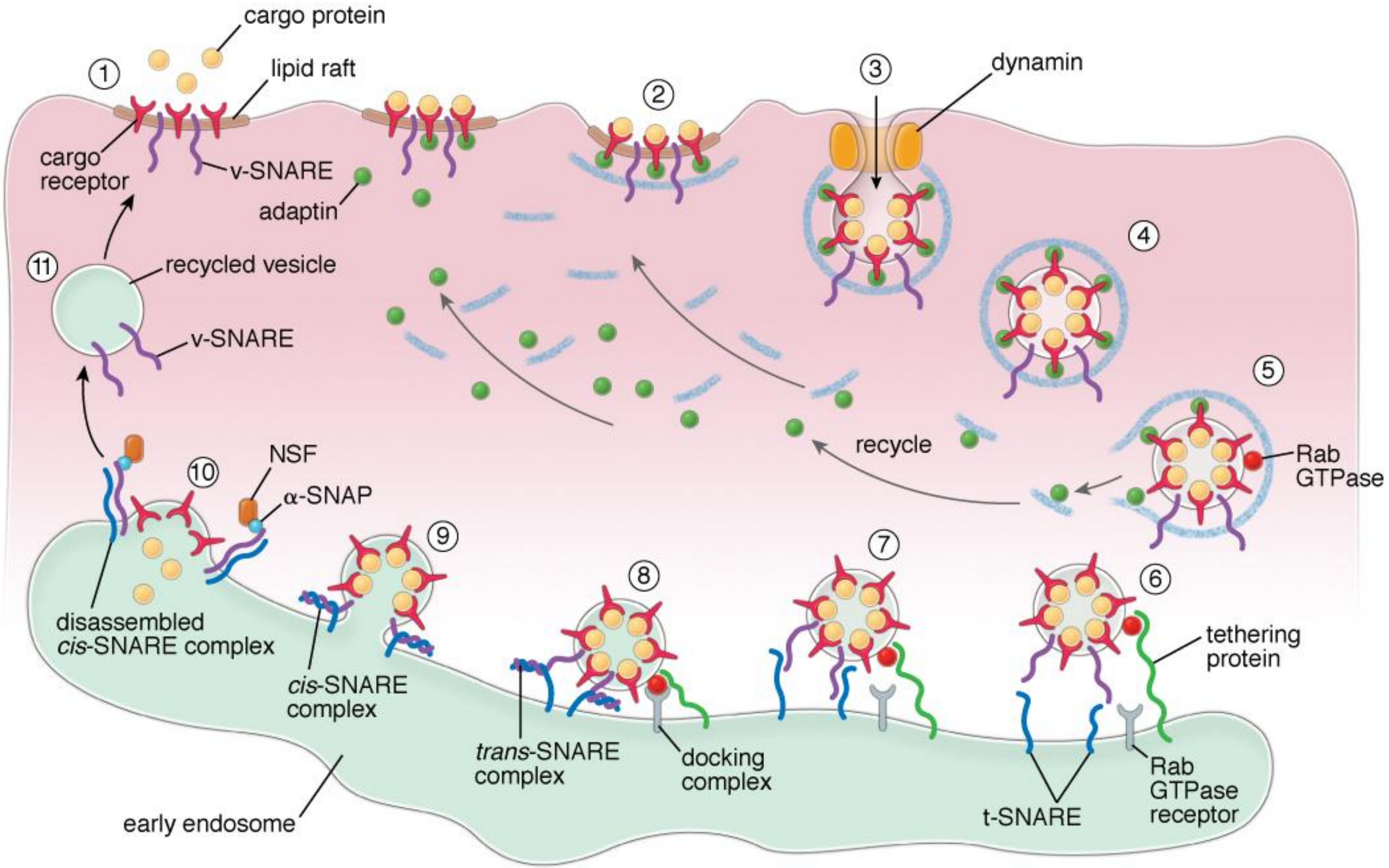
Regulation mechanisms and proteins

- Vesicular transport requires careful regulation by proteins and adhesion molecules



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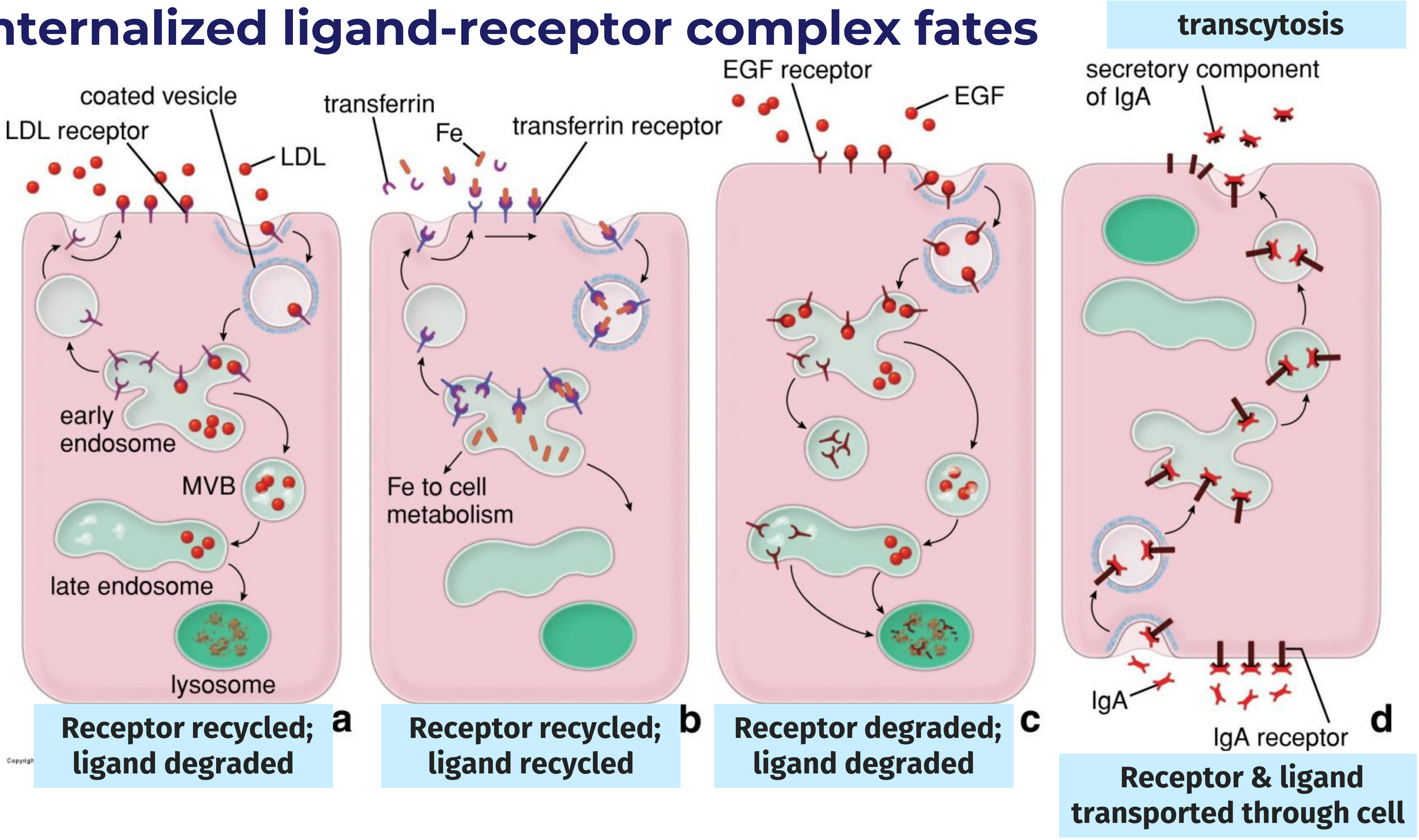
Internalized ligand-receptor complex fates



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Practice Question

- Which of the following best characterizes vesicular transport?

a) Movement of proteins within a cell
b) Movement of proteins from inside to outside of a cell
c) Movement of proteins from outside to inside of a cell
d) All of the above

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Practice Question

- Which of the following best characterizes the vesicular transport of iron (Fe) into a cell?

a) The receptor is recycled and the ligand degraded

b) Both ligand and receptor pass through the cell without degradation

c) Both receptor and ligand are recycled

d) Both receptor and ligand are degraded

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